

TradingAI Software and Component Inventory

Complete current runtime, development, external-service, UI and internal-module inventory.

System reference date: 2026-08-29

Standalone current-state reference for the complete platform through Milestone 78.

1. Platform runtime

Layer	Current software / authority	Role in TradingAI
Host	macOS	Primary local production/paper-trading workstation; launchd/LaunchAgents and Homebrew services.
Backend language	Python >= 3.13	Ingestion, intelligence, research, risk, portfolio, execution, management, API and operations.
Python environment	uv	Virtual environment, dependency resolution and command execution.
Database	PostgreSQL 17	Authoritative relational state, snapshots, publications, lineage, broker/portfolio/execution records.
ORM / migrations	SQLAlchemy 2.x + Alembic	Data access and schema evolution.
API	FastAPI + Uvicorn	Production HTTP API consumed by the workstation and operational clients.
Frontend	React 18 + TypeScript + Vite	Institutional workstation UI.
Cache / runtime infra	Redis	Configured caching/runtime dependency where used.
Scheduling	macOS launchd + APScheduler	OS-level recurring services and application scheduling.
Market data	Polygon	Current authoritative provider for underlying/options market data used by the platform.
Broker	Interactive Brokers Paper via TWS/Gateway	Order, fill, account and broker portfolio truth.
Earnings	Alpha Vantage	Corporate earnings calendar authority.
Macro calendar	Federal Reserve, BLS, BEA	Primary-source macro event authorities.

2. Python dependency set

The current pyproject.toml defines the following runtime and development packages.

- numpy
- pandas
- scipy
- sqlalchemy>=2.0
- psycpg2-binary
- python-dotenv
- pydantic>=2.0
- pydantic-settings
- loguru
- requests
- httpx
- polygon-api-client>=1.16.3
- plotly
- streamlit
- ta
- apscheduler
- redis
- xgboost
- scikit-learn
- alembic
- matplotlib>=3.11.0

- certifi>=2025.11.12
- fastapi>=0.139.2
- uvicorn>=0.49.0
- pytest
- pytest-cov
- ruff
- mypy
- black>=26.5.1
- mypy>=2.1.0
- pytest>=9.1.1
- ruff>=0.15.20

3. Frontend dependency set

- @types/node ^22.10.0
- @types/react ^18.3.18
- @types/react-dom ^18.3.5
- @vitejs/plugin-react ^4.3.4
- lucide-react ^0.468.0
- react ^18.3.1
- react-dom ^18.3.1
- recharts ^2.15.0
- typescript ^5.7.2
- vite ^6.0.5

4. Internal backend application components

The backend is intentionally modular. The following top-level packages exist in the current source tree or are added by the M78 release overlay.

Package	Current responsibility
src/trading_ai/advanced_trade_builder	Exact multi-leg trade construction, validation, entry/stop/target and management-plan assembly.
src/trading_ai/analytics_dashboard	Analytical dashboard services and reporting.
src/trading_ai/app	App subsystem.
src/trading_ai/authoritative_paper_trading	Governed paper-trading authority and lifecycle state.
src/trading_ai/autonomous_position_management	Automated management of filled positions.
src/trading_ai/backtest	Backtest subsystem.
src/trading_ai/broker	Broker integration abstractions and IBKR implementation.
src/trading_ai/broker_portfolio_sync	Broker-to-database portfolio/account synchronization.
src/trading_ai/cache	Cache subsystem.
src/trading_ai/certification	Certification contracts and governed promotion controls.
src/trading_ai/config	Config subsystem.
src/trading_ai/core	Core subsystem.
src/trading_ai/daily	Daily subsystem.
src/trading_ai/daily_scan_workstation	Daily Scan Workstation subsystem.
src/trading_ai/database	SQLAlchemy sessions, models, repositories and persistence utilities.
src/trading_ai/decision	Decision subsystem.
src/trading_ai/deployment	Deployment subsystem.
src/trading_ai/domain	Domain subsystem.
src/trading_ai/downside_risk_veto	Downside Risk Veto subsystem.
src/trading_ai/dynamic_position_management	Ongoing exit/management decisions for open positions.
src/trading_ai/execution	Execution subsystem.
src/trading_ai/execution_intelligence	Executable-price, slippage and fill-quality intelligence.
src/trading_ai/execution_orchestration	Order-intent orchestration and execution sequencing.
src/trading_ai/execution_workspace	Execution-workspace services and operator-facing execution state.
src/trading_ai/feature_store	Feature Store subsystem.
src/trading_ai/futures_intelligence	Futures context and pre-open intelligence.
src/trading_ai/historical_underlying_replay	Point-in-time historical underlying replay used by M77 research.
src/trading_ai/indicators	Indicators subsystem.
src/trading_ai/inflection_intelligence	Governed multi-factor inflection detection and timing intelligence.
src/trading_ai/institutional_intelligence	Integrated institutional decision/intelligence layer.
src/trading_ai/institutional_market_structure	Institutional market-structure evidence.
src/trading_ai/institutional_options	Underlying-first institutional options strategy generation and optimization.
src/trading_ai/lifecycle_governance	Lifecycle state and governance controls.
src/trading_ai/lineage	Authority provenance and lineage validation.
src/trading_ai/live_trading_governance	Live-trading certification, environment isolation, controls and audit.
src/trading_ai/market	Underlying/index instruments, providers and ingestion support.
src/trading_ai/market_intelligence	Correlation, sentiment, breadth, sector and cross-asset intelligence.
src/trading_ai/market_overview	Current market context/health/regime dashboard authority.
src/trading_ai/models	Models subsystem.
src/trading_ai/observability	Operational telemetry and health reporting.
src/trading_ai/operational_resilience	Recovery, replay and resilience controls.

Package	Current responsibility
src/trading_ai/opex_intelligence	OPEX/futures-derived market context.
src/trading_ai/opportunity_domain	Canonical opportunity lifecycle and state model.
src/trading_ai/optimization	Optimization subsystem.
src/trading_ai/option_valuation_intelligence	Option fair value, surface, event and relative-value intelligence.
src/trading_ai/options	Options provider, chain, quote and analytics primitives.
src/trading_ai/order_management	Order-state management and lifecycle reconciliation.
src/trading_ai/outcome_probability	Outcome-probability modeling/calibration components.
src/trading_ai/paper_trading	Paper-trading scheduler, automation control plane, recovery and observability.
src/trading_ai/performance_learning	Prediction-vs-actual, attribution and learning analytics.
src/trading_ai/polygon_intelligence	Polygon-specific acquisition/intelligence helpers.
src/trading_ai/portfolio	Portfolio subsystem.
src/trading_ai/portfolio_intelligence	Portfolio risk, fit, allocation and ranking services.
src/trading_ai/portfolio_management	Portfolio Management subsystem.
src/trading_ai/portfolio_risk_allocation	Risk snapshots, optimization and allocation publications.
src/trading_ai/portfolio_risk_management	Portfolio Risk Management subsystem.
src/trading_ai/position_management	Position Management subsystem.
src/trading_ai/position_monitoring	Position Monitoring subsystem.
src/trading_ai/production_api	Production Api subsystem.
src/trading_ai/production_operations	Production Operations subsystem.
src/trading_ai/providers	Providers subsystem.
src/trading_ai/published_state	Published State subsystem.
src/trading_ai/ranking	Ranking subsystem.
src/trading_ai/realtime_monitoring	Realtime Monitoring subsystem.
src/trading_ai/replay	Replay subsystem.
src/trading_ai/reporting	Reporting subsystem.
src/trading_ai/repositories	Repositories subsystem.
src/trading_ai/research	Research subsystem.
src/trading_ai/research_workstation	Research Workstation subsystem.
src/trading_ai/risk	Risk subsystem.
src/trading_ai/risk_gateway	Risk Gateway subsystem.
src/trading_ai/scanner	Scanner subsystem.
src/trading_ai/services	Services subsystem.
src/trading_ai/stock_intelligence	Stock Intelligence subsystem.
src/trading_ai/strategy	Strategy subsystem.
src/trading_ai/strategy_engine	Strategy Engine subsystem.
src/trading_ai/trade_plan_certification	Trade Plan Certification subsystem.
src/trading_ai/trend_intelligence	Multi-timeframe trend state, transition, forecasting and participation/leadership intelligence.
src/trading_ai/ui	Ui subsystem.
src/trading_ai/ui_backup_simple	Ui Backup Simple subsystem.
src/trading_ai/universe	Universe subsystem.
src/trading_ai/utls	Utils subsystem.
src/trading_ai/volatility	Volatility subsystem.
src/trading_ai/walkforward	Walkforward subsystem.
src/trading_ai/setup_intelligence	M78 shadow-only setup taxonomy, lifecycle, outcomes, probability, EV, option expression, ranking and certification.

5. M78 Setup Intelligence software

Milestone 78 is an additive shadow-only subsystem. It does not replace any production source module.

Component	Role
contracts.py	Setup taxonomy, lifecycle stages, directions, statuses and shared contracts.
detector.py	Point-in-time setup detection and lifecycle continuity.
policy.py	Readiness, model and governance policy.
models.py	Persistence models/contracts for M78 entities.
repository.py	Database access for snapshots, transitions, outcomes, models, predictions, publications, certifications and audit.
probability.py	Hierarchical empirical probability, sample gates and shrinkage.
expected_value.py	Expected return/R and capital/time efficiency.
option_expression.py	Shadow-only setup-aware option-expression guidance.
service.py	Capture, outcome materialization, training, approval, activation, prediction, status and certification orchestration.
scripts/run_m78_setup_intelligence.py	Operator CLI.
scripts/run_m78_daily_shadow.py	Daily capture/outcome/prediction orchestrator.

6. Operational services and current schedules

Service class	Examples	Current scheduling rule
Weekday ingestion	M71 futures pre-open; M69.6 event intelligence; morning ingestion; intraday options; EOD ingestion	Monday-Friday only; original times preserved.
Nightly research	Combined M77/M78 shadow LaunchAgent	Monday-Friday 18:30; M78 gated by explicit M77 READY.
Continuous operations	Broker portfolio sync; production operations; dynamic management; portfolio intelligence; entry-fill management	Remain continuously managed; not converted to weekday calendar jobs.

7. Development and validation tooling

- pytest / pytest-cov for focused and regression tests
- ruff and black for formatting/linting
- mypy for static typing checks
- Alembic for versioned database migrations
- py_compile for installation-time syntax validation
- shell installers with timestamped backups and rollback scripts
- launchctl/plutil for user LaunchAgent registration and validation